

Assignment 2. Data Collection & Understanding

Due Date: December 7, 2024

Evaluation: 10% of the total grade

Instructions

For this assignment, extend your work performed on dataset(s) from the previous assignment. The data understanding phase of CRISP-DM involves taking a closer look at the data available for mining. You can use one or several datasets related to the problem you are solving.

1. Decide and confirm on the data to be used for analysis. Criteria include relevance to the data mining goals, quality, and technical constraints such as limits on data volume or data types. Make sure you have access to the data you want to use in the quantity and quality you need. Many online data sources strongly throttle the data volume you can get through their API. Others have a higher limit but have use restrictions that you must respect.
2. Describe data collection process. List the dataset(s) acquired, together with their locations, the methods used to acquire them, and any problems encountered. Record problems encountered and any resolutions achieved.
3. Describe the data that has been acquired, including the format of the data, the quantity of data (for example, the number of records and fields in each table), the identities of the fields, and any other surface features which have been discovered.
4. Explore data. This task addresses data mining questions using visualization and reporting techniques. These include distribution of key attributes (for example, the target attribute of a prediction task) relationships between pairs or small numbers of attributes, results of simple aggregations, properties of significant sub-populations, and simple statistical analyses. These analyses may directly address the data mining goals; they may also contribute to or refine the data description and quality reports, and feed into the transformation and other data preparation steps needed for further analysis.

Use Jupyter notebook for this assignment and include code, text, your comments, observations, and visualizations. Please make sure that the submitted notebooks have been run and the cell outputs are visible. Once created, submit it in pdf format.

Scoring Rubric

Category	Criteria	Marks
Data Collection	Data collection procedure explained fully, including a) Source of data identified clearly (website address, Api, etc.) b) Methods used to acquire data	2
Descriptive Statistics	Descriptive statistics is given for every dataset and variable used in analysis.	2
Data Exploration	Statistical analysis is performed. Insightful discussion of results is provided.	4
Code	Python code is exceptionally well organized and easy to follow.	2
Total		10